

# James C. Davis

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## RESEARCH THEME

My research develops empirical and engineering foundations for trustworthy software systems. I use evidence from experiments, mining software repositories, and real deployments to guide the design, verification, and security of software and its supply chains. By integrating empirical study with practical tool building, my group strengthens the reliability and resilience of the software that underpins modern computing.

## EDUCATION

|                                                                                             |           |
|---------------------------------------------------------------------------------------------|-----------|
| <b>Ph.D, Computer Science and Applications</b><br><i>Virginia Tech, Blacksburg, VA</i>      | 2015–2020 |
| <b>B.Sc. Computer Science, B.Sc. Mathematics</b><br><i>Clarkson University, Potsdam, NY</i> | 2008–2012 |

## PROFESSIONAL EXPERIENCE

|                                                                                                                                          |                   |
|------------------------------------------------------------------------------------------------------------------------------------------|-------------------|
| <b>Assistant Professor</b><br><i>Purdue University — Electrical and Computer Engineering</i>                                             | Fall 2020–present |
| <b>Intern, Microsoft Research (RiSE group: Cloud Security)</b><br><i>Microsoft Research, Redmond, WA — Mentored by Patrice Godefroid</i> | Summer 2019       |
| <b>Intern, IBM Research (Storage)</b><br><i>IBM Research, Almaden, CA — Mentored by Deepavali Bhagwat</i>                                | Summer 2018       |
| <b>Graduate Research Assistant</b><br><i>Virginia Tech — Advised by Dongyoon Lee</i>                                                     | 2016–2020         |
| <b>Software Engineer, IBM (GPFS)</b><br><i>IBM, Poughkeepsie, NY</i>                                                                     | 2012–2017         |

## EXTERNAL GRANTS

**TOTAL: \$5,262,934. TOTAL AS PI: \$2,576,026. MY TOTAL SHARE: \$3,391,251.<sup>1</sup>**

|                                                                                                               |
|---------------------------------------------------------------------------------------------------------------|
| [G-1] NSF #2541917: CAREER: PTM-SEER: Software Engineering Foundations for Re-Using Pre-Trained Neural Models |
| PI                                                                                                            |
| <i>US National Science Foundation</i>                                                                         |
| <i>2026–2031. \$687,140; my share: \$687,140.</i>                                                             |
| Other personnel: Sole PI                                                                                      |

<sup>1</sup>Calculation: *Total* is the sum of all awards to Purdue on which I have been PI or Co-PI. *Total as PI* is the sum of all awards to Purdue on which I have been the PI — not all of these funds are managed by me. *My total share* is the sum of funds I have managed across all awards.

- [G-2] **NSF #2526621: Collaborative Research: SaTC 2.0: RES: AIGIS: Securing the Deep Learning Model Supply Chain**  
Purdue PI (Columbia University is lead)  
*US National Science Foundation*  
*2026–2030. Total: \$1,220,000; Purdue: \$410,000; my share: \$410,000.*  
Other personnel: NSF #2526620: Junfeng Yang (Computer Science, Columbia University); NSF #2526622: Vasileios Kemerlis (Computer Science, Brown University)
- [G-3] **NSF #2504445: OAC Core: Cyberinfrastructure for Multi-Stream Architectures Applied to Computer Vision: Efficiency via Co-Design of Network Architectures and Framework Operators**  
Co-PI  
*US National Science Foundation*  
*2025–2028. \$660,000; my share: \$330,000.*  
Other personnel: PI: Yung-Hsiang Lu (Electrical & Computer Engineering)
- [G-4] **NSF #2452533: RFE: Research: Developing and Piloting a Prompt Engineering Competency Framework for Software Engineering Education**  
Co-PI  
*US National Science Foundation*  
*2025–2028. \$350,000; my share: \$65,716.*  
Other personnel: PI: Kirsten Davis (Engineering Education)
- [G-5] **NSF #2537308: Collaborative Research: Planning: CROSS: Building a Community aROund Securing the Research Software Supply Chain**  
PI (Purdue University is lead)  
*US National Science Foundation*  
*2025–2027. Total: \$200,000; Purdue: \$105,113; my share: \$82,213.*  
Other personnel: Co-PI: Alexandra Harris-Watson (Psychological Sciences); NSF #2537310: Jeffrey C. Carver (Computer Science, University of Alabama Tuscaloosa); NSF #2537309: George K. Thiruvathukal (Computer Science, Loyola University Chicago)
- [G-6] **Rolls Royce: Securing Software Implementations through System Fuzz Testing and Modular Formal Methods**  
PI  
*Contract with Rolls Royce*  
*2025–2026. \$181,447; my share: \$90,723.5.*  
Other personnel: Co-PI: Aravind Machiry (Electrical & Computer Engineering)
- [G-7] **Qualcomm Innovation Fellowship**  
PI  
*Qualcomm, Inc.*  
*2025–2026. \$100,000; my share: \$100,000.*  
Other personnel: Project Supervisor of winning team: P. Amusuo, D. Anandayuvraj

- [G-8] **Unrestricted Gift: Typosquat Detection in Open-Source Ecosystems**  
PI  
*Socket, Inc.*  
2025. \$20,000; my share: \$20,000.  
Other personnel: Sole PI
- [G-9] **AutoUP: Automated Unit Proofing**  
PI  
*OpenAI — Cybersecurity Grant Program*  
2025. \$5,000 (API credits); my share: \$5,000 (API credits).  
Other personnel: Sole PI
- [G-10] **NSF #2343596: Collaborative Research: EAGER: CET: Exploring The Risks and Rewards of Large Language Models in Enabling Energy-Efficient Data Center Software Infrastructure**  
Purdue PI (Loyola University Chicago is lead)  
*US National Science Foundation*  
2024–2026. Total: \$297,923; Purdue: \$149,976; my share: \$74,988.  
Other personnel: Co-PI: Yung-Hsiang Lu (Electrical & Computer Engineering); NSF #2343595: George K. Thiruvathukal (Computer Science, Loyola University Chicago)
- [G-11] **Rolls Royce: Facilitating Effective Dynamic Analysis of Embedded Software**  
Co-PI  
*Contract with Rolls Royce*  
2024–2025. \$150,000; my share: \$75,000.  
Other personnel: PI: Aravind Machiry (Electrical & Computer Engineering)
- [G-12] **NSF #2512797: Travel: NSF Student Travel Grant for 2025 International Conference on Software Engineering**  
PI  
*US National Science Foundation*  
2024–2025. \$24,000; my share: \$24,000.  
Other personnel: Sole PI
- [G-13] **Unrestricted Gift: Improving OSS Supply Chain Security by Promoting Software Signing**  
Co-PI  
*Google, LLC*  
2023. \$200,000; my share: \$100,000.  
Other personnel: PI: Santiago Torres-Arias (Electrical & Computer Engineering)
- [G-14] **Efficient Computer Vision for Edge Devices**  
Co-PI  
*Contract with Cisco*  
2023–2024. \$179,941; my share: \$89,970.5.  
Other personnel: PI: Yung-Hsiang Lu (Electrical & Computer Engineering)

- [G-15] **Rolls Royce: Dynamic Security Analysis of Embedded Software Systems**  
Co-PI  
*Contract with Rolls Royce*  
*2023-2024. \$150,000; my share: \$75,000.*  
Other personnel: PI: Aravind Machiry (Electrical & Computer Engineering)
- [G-16] **NSF #2135156: Collaborative Research: SaTC: CORE: Small: Improving Sanitization and Avoiding Denial of Service Through Correct and Safe Regexes**  
PI (Purdue University is lead)  
*US National Science Foundation*  
*2022-2025. Total: \$548,000; Purdue: \$274,000; my share: \$274,000.*  
Other personnel: NSF #2135157: Dongyoon Lee (Computer Science, SUNY at Stony Brook)
- [G-17] **NSF #2229703: POSE: Phase I: Scoping An Open-Source Ecosystem Around Proactive Software Supply Chain Monitoring**  
Co-PI  
*US National Science Foundation*  
*2022-2023. \$300,000; my share: \$150,000.*  
Other personnel: PI: Santiago Torres-Arias (Electrical & Computer Engineering)
- [G-18] **Cisco: Monitor and manage security risks in software supply chains with Sigstore**  
Co-PI  
*Contract with Cisco*  
*2022-2023. \$184,536; my share: \$92,268.*  
Other personnel: PI: Santiago Torres-Arias (Electrical & Computer Engineering)
- [G-19] **Cisco: Trustworthy Re-use of Pre-Trained Neural Networks**  
PI  
*Contract with Cisco*  
*2022-2023. \$179,237; my share: \$89,618.5.*  
Other personnel: Co-PI: Yung-Hsiang Lu (Electrical & Computer Engineering)
- [G-20] **Unrestricted Gift: Machine Learning Reproducibility**  
PI  
*Google, LLC*  
*2022. \$80,000; my share: \$80,000.*  
Other personnel: Sole PI
- [G-21] **NSF #2107230: Collaborative Research: OAC Core: Advancing Low-Power Computer Vision at the Edge**  
Co-PI (Purdue University is lead)  
*US National Science Foundation*  
*2021-2024. Total: \$508,000; Purdue: \$258,000; my share: \$129,000.*  
Other personnel: PI: Yung-Hsiang Lu (Electrical & Computer Engineering); NSF #2107020: George K. Thiruvathukal (Computer Science, Loyola University Chicago)

[G-22] **Rolls Royce: Dynamic Analysis of Embedded Firmware**

Co-PI

*Contract with Rolls Royce*

*2021–2022. \$175,000; my share: \$87,500.*

Other personnel: PI: Aravind Machiry (Electrical & Computer Engineering)

[G-23] **Unrestricted Gift: Research on Machine Learning Reproducibility**

Co-PI

*Google, LLC*

*2020. \$100,000 (\$80,000 + \$20,000); my share: \$100,000 (\$80,000 + \$20,000).*

Other personnel: PI: Yung-Hsiang Lu (Electrical & Computer Engineering)

**INTERNAL GRANTS**

***TOTAL: \$234,431.***

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[IG-1] **Exploring the Impact and the Use of Generative Models in Computer Engineering Education**

Co-PI

*Office of the Provost, through the program “AI in teaching and learning grants”*

*2023-2024. \$79,431; my share: \$79,431.*

Other personnel: PI: Aravind Machiry (Electrical & Computer Engineering); Co-PI: Carla Zoltowski (Electrical & Computer Engineering); Co-PI: Justin Hess (Engineering Education); Co-PI: Yung-Hsiang Lu (Electrical & Computer Engineering)

[IG-2] **Revamping the CompE Curriculum for Secure Software Engineering**

PI

*ECE Agile Reform of Curriculum program, enabled by Elmore Family gift*

*2021–2022. \$150,000; my share: \$150,000.*

Other personnel: Co-PI: Aravind Machiry (Electrical & Computer Engineering); Co-PI: Santiago Torres-Arias (Electrical & Computer Engineering); Co-PI: Saurabh Bagchi (Electrical & Computer Engineering)

[IG-3] **Intercultural Engineering Education for Software Engineers**

PI

*Purdue University VEIL Program*

*2020. \$5,000; my share: \$5,000.*

Other personnel: Co-PI: Kirsten Davis (Engineering Education)

## GIFTS IN KIND

**TOTAL VALUATION: \$16,478.**

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[IG-1] **Intel: Gift of Laptops for Educational Research on Local LLMs for Education**

PI

*Gift in Kind (Equipment)*

2026. \$16,478; my share: \$16,478.

Other personnel: Co-PI: Sean Brophy (Engineering Education)

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## REFEREED CONFERENCE PUBLICATIONS (FULL PAPERS)<sup>2</sup> *These venues are CORE2026 rank A or A\*.<sup>3</sup>*

- [C-1] P.C. Amusuo, R. Calvo, D. Anandayuvraj, T. Le Lievre, K. Kolyakov, E. Jorgensen, A. Machiry, and **J.C. Davis**. *AutoSOUP: Safety-Oriented Unit Proof Generation for Component-level Memory-Safety Verification*. Proceedings of the 33rd ACM Conference on Computer and Communications Security (CCS'26). 15 pages.
- [C-2] C. Okafor, **J.C. Davis**, and S. Torres-Arias. *Context-Aware Trust Verification for Identity-Based Software Signing*. Proceedings of the 41st IEEE/ACM International Conference on Automated Software Engineering (ASE'26). 21.2% acceptance rate (265/1,236). 13 pages.
- [C-3] K.G. Kalu, S. Okorafor, T. Singla, S. Chen, S. Torres-Arias, and **J.C. Davis**. *Why Johnny Adopts Identity-Based Software Signing: A Usability Case Study of Sigstore*. Proceedings of the 35th USENIX Security Symposium (USENIX Security'26). 13.2% acceptance rate (364/2,754). 20 pages.
- [C-4] Y. Liu, B. Çakar, A. Agrawal, M. Seo, **J.C. Davis**, and Dongyoon Lee. *Regular Expression Denial of Service Induced by Backreferences*. Proceedings of the 35th USENIX Security Symposium (USENIX Security'26). 13.2% acceptance rate (364/2,754). 20 pages.
- [C-5] P. Jajal, N.J. Eliopoulos, B.S.H. Chou, G.K. Thiruvathukal, Y.H. Lu, and **J.C. Davis**. *AdaPerceiver: Transformers with Adaptive Width, Depth, and Tokens*. The IEEE/CVF Conference on Computer Vision and Pattern Recognition — Findings Track (2026). 35.3% acceptance rate ((4,090+1,600)/16,092). 11 pages.
- [C-6] P. Jajal\*, N. Eliopoulos\*, B.S.H. Chou, G.K. Thiruvathukal, **J.C. Davis**, and Y.H. Lu. *Inference-Time Alignment of Diffusion Models with Evolutionary Algorithms*. The IEEE/CVF Conference on Computer Vision and Pattern Recognition — Findings Track (2026). 35.3% acceptance rate ((4,090+1,600)/16,092). 12 pages.
- [C-7] A. Rozema and **J.C. Davis**. *Anti-Phishing Training (Still) Does Not Work: A Reproduction of Phishing Training Inefficacy Grounded in the NIST Phish Scale*. Proceedings of the ACM Web Conference 2026 (WWW'26). 20.1% acceptance rate (676/3,370). 12 pages.
- [C-8] W. Jiang, B. Çakar, M. Lysenko, and **J.C. Davis**. *ConfuGuard: Using Metadata to Detect Active and Stealthy Package Confusion Attacks Accurately and at Scale*. Proceedings of the 48th International Conference on Software Engineering (ICSE'26). 21.8% acceptance rate (321/1,469). 13 pages.

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<sup>2</sup>Here and elsewhere, my research mentees are underlined. These are students whose work I supervised during the research project. My name is given in **bold**. In these publication venues, the final author (“anchor author”) provides the primary intellectual supervision of the work.

<sup>3</sup>In my research areas, top-tier conferences are the most prestigious venues for publication, with some journals considered comparable. The ICORE rankings are an international ranking system of conferences in which A\* venues are “best” and A venues are “reputable” venues; for either category, the peer reviewers are recognized experts in the field. Acceptance rates of ~10–25% across 100+ submissions are a good indicator of conference quality.

- [C-9] D. Anandayuvraj, T. Singla, Z. Hammadeh, A. Lund, A. Holloway, and **J.C. Davis**. *Learning From Software Failures: A Case Study at a National Space Research Center*. Proceedings of the 48th International Conference on Software Engineering (ICSE'26). 21.8% acceptance rate (321/1,469). 13 pages.
- [C-10] P.C. Amusuo, O. Cochell, T. Le Lievre, P.V. Patil, A. Machiry, and **J.C. Davis**. *Do Unit Proofs Work? An Empirical Study of Compositional Bounded Model Checking for Memory Safety Verification*. Proceedings of the 48th International Conference on Software Engineering (ICSE'26). 21.8% acceptance rate (321/1,469). 13 pages.
- [C-11] B.S.H. Chou, P. Jajal, N.J. Eliopoulos, **J.C. Davis**, G.K. Thiruvathukal, K.Y.J. Yun, and Y.H. Lu. *LadderSym: A Multimodal Interleaved Transformer for Music Practice Error Detection*. Proceedings of the Fourteenth International Conference on Learning Representations (ICLR'26). 28% acceptance rate (5,000/18,000). 22 pages.
- [C-12] A. Kellas, N. Christou, W. Jiang, P. Li, L. Simon, Y. David, V. Kemerlis, **J.C. Davis**, and J. Yang. *PickleBall: Secure Deserialization of Pickle-based Machine Learning Models*. Proceedings of the 32nd ACM Conference on Computer and Communications Security (CCS'25). 13.9% acceptance rate (316/2,278). 16 pages. *Best Artifact Award*.
- [C-13] P.C. Amusuo, K.A. Robinson, T. Singla, H. Peng, A. Machiry, S. Torres-Arias, L. Simon, and **J.C. Davis**. *ZTD<sub>JAVA</sub>: Mitigating Software Supply Chain Vulnerabilities via Zero-Trust Dependencies*. Proceedings of the 47th International Conference on Software Engineering (ICSE'25). 24% acceptance rate (248/1031). 13 pages.
- [C-14] B.S.H. Chou, P. Jajal, N.J. Eliopoulos, T. Nadolsky, C.Y. Yang, N. Ravi, **J.C. Davis**, K.Y.J. Yun, and Y.H. Lu. *A Musician's Muse: Detecting Performance Errors with Transformers*. Proceedings of the 39th Annual AAAI Conference on Artificial Intelligence (AAAI'25). 23% acceptance rate (3,032/12,957). 11 pages.
- [C-15] S.R. Tanksalkar, S. Muralee, D.M.S.H. Danduri, P. Amusuo, A. Bianchi, **J.C. Davis**, and A. Machiry. *LEMIX: Enabling Testing of Embedded Applications as Linux Applications*. Proceedings of the 34th USENIX Security Symposium (USENIX Security'25). 17.1% acceptance rate (407/2,385). 30 pages.
- [C-16] K. Kalu, T. Singla, C. Okafor, S. Torres-Arias, and **J.C. Davis**. *An Industry Interview Study of Software Signing for Supply Chain Security*. Proceedings of the 34th USENIX Security Symposium (USENIX Security'25). 17.1% acceptance rate (407/2,385). 18 pages.
- [C-17] M. Shen, A. Pillai, B.A. Yuan, **J.C. Davis**, and A. Machiry. *Finding 709 Defects in 258 Projects: An Experience Report on Applying CodeQL to Open-Source Embedded Software (Experience Paper)*. Proceedings of the 34th ACM SIGSOFT International Symposium on Software Testing and Analysis (ISSTA'25). 21% acceptance rate (115/553). 22 pages.
- [C-18] M.H.M. Bhuiyan\*, B. Çakar\*, E. Burmane, **J.C. Davis**, and C.A. Staicu. *SoK: A Literature and Engineering Review of Regular Expression Denial of Service*. Proceedings of the 20th ACM ASIA Conference on Computer and Communications Security (AsiaCCS) Cycle 2 (AsiaCCS'25). 20% acceptance rate (53/269). 22 pages.
- [C-19] N. Eliopoulos, P. Jajal, **J.C. Davis**, G. Liu, G.K. Thiruvathukal, and Y.H. Lu. *Pruning One More Token is Enough: Leveraging Latency-Workload Non-Linearities for Vision Transformers on the Edge*. Proceedings of the IEEE/CVF Winter Conference on Applications of Computer Vision (WACV'25). 37.8% acceptance rate (930/2,458). 12 pages.

- [C-20] P. Jajal, N. Eliopoulos, B. Chou, G.K. Thiruvathukal, **J.C. Davis**, and Y.H. Lu. *Token Turing Machines are Efficient Vision Models*. Proceedings of the IEEE/CVF Winter Conference on Applications of Computer Vision (WACV'25). 37.8% acceptance rate (930/2,458). 12 pages.
- [C-21] D. Anandayuvraj, M. Campbell, A. Tewari, and **J.C. Davis**. *FAIL: Analyzing Software Failures from the News Using LLMs*. Proceedings of the 38th IEEE/ACM International Conference on Automated Software Engineering (ASE'24). 26% acceptance rate (154/587). 13 pages.
- [C-22] P. Jajal, W. Jiang, A. Tewari, E. Kocinare, J. Woo, A. Sarraf, Y.H. Lu, G.K. Thiruvathukal, and **J.C. Davis**. *Interoperability in Deep Learning: A User Survey and Failure Analysis of ONNX Model Converters*. Proceedings of the 33rd ACM SIGSOFT International Symposium on Software Testing and Analysis (ISSTA'24). 21% acceptance rate (143/694). 13 pages.
- [C-23] J. Jones, W. Jiang, N. Synovic, G.K. Thiruvathukal, and **J.C. Davis**. *What do we know about Hugging Face? A systematic literature review and quantitative validation of qualitative claims*. Proceedings of the 18th ACM/IEEE International Symposium on Empirical Software Engineering and Measurement (ESEM'24). 24% acceptance rate (34/139). 12 pages.
- [C-24] L. Franke, H. Liang, S. Farzanehpour, A. Brantly, **J.C. Davis**, and C. Brown. *An Exploratory Mixed-methods Study on General Data Protection Regulation (GDPR) Compliance in Open-Source Software*. Proceedings of the 18th ACM/IEEE International Symposium on Empirical Software Engineering and Measurement (ESEM'24). 24% acceptance rate (34/139). 12 pages.
- [C-25] J. Chen, D. Anandayuvraj, **J.C. Davis**, and S. Rahaman. *On the Contents and Utility of IoT Cybersecurity Guidelines*. Proceedings of the ACM on Software Engineering (PACMSE), Issue FSE 2024 (FSE'24). 26% acceptance rate (121/474). 24 pages.
- [C-26] T.R. Schorlemmer, K.G. Kalu, L. Chigges, K.M. Ko, E.A.M.A. Ishgair, S. Bagchi, S. Torres-Arias, and **J.C. Davis**. *Signing in Four Public Software Package Registries: Quantity, Quality, and Influencing Factors*. Proceedings of the 45th IEEE Symposium on Security and Privacy (IEEE S&P'24). 18% acceptance rate (261/1463). 16 pages.
- [C-27] W. Maxam and **J.C. Davis**. *An Interview Study on Third-Party Cyber Threat Hunting Processes in the U.S. Department of Homeland Security*. Proceedings of the 33rd USENIX Security Symposium (USENIX Security'24). 18% acceptance rate (382/2176). 18 pages.
- [C-28] W. Jiang, J. Yasmin, J. Jones, N. Synovic, J. Kuo, N. Bielanski, Y. Tian, G.K. Thiruvathukal, and **J.C. Davis**. *PeaTMOSS: A Dataset and Initial Analysis of Pre-Trained Models in Open-Source Software*. Proceedings of the 21st Annual Conference on Mining Software Repositories (MSR'24). 29% acceptance rate (42/146). 13 pages.
- [C-29] W. Jiang, N. Synovic, M. Hyatt, T.R. Schorlemmer, R. Sethi, Y.H. Lu, G.K. Thiruvathukal, and **J.C. Davis**. *An Empirical Study of Pre-Trained Model Reuse in the Hugging Face Deep Learning Model Registry*. Proceedings of the ACM/IEEE 45th International Conference on Software Engineering (ICSE'23). 26% acceptance rate (208/796). 13 pages.
- [C-30] P.C. Amusuo, R.A.C. Méndez, Z. Xu, A. Machiry, and **J.C. Davis**. *Systematically Detecting Packet Validation Vulnerabilities in Embedded Network Stacks*. Proceedings of the 38th IEEE/ACM International Conference on Automated Software Engineering (ASE'23). 21% acceptance rate (134/629). 13 pages.
- [C-31] S.A. Hassan, Z. Aamir, D. Lee, **J.C. Davis**, and F. Servant. *Improving Developers' Understanding of Regex Denial of Service Tools through Anti-Patterns and Fix Strategies*. Proceedings of the 44th IEEE Symposium on Security and Privacy (IEEE S&P'23). 17.1% acceptance rate (196/1,147). 18 pages.

- [C-32] A. Goel, C. Tung, N. Eliopoulos, X. Hu, G.K. Thiruvathukal, **J.C. Davis**, and Y.H. Lu. *Directed Acyclic Graph-based Neural Networks for Tunable Low-Power Computer Vision*. Proceedings of the ACM/IEEE International Symposium on Low Power Electronics and Design (ISLPED'22). 24% acceptance rate (23/96). 6 pages.
- [C-33] E. Barlas, X. Du, and **J.C. Davis**. *Exploiting Input Sanitization for Regex Denial of Service*. Proceedings of the ACM/IEEE 44th International Conference on Software Engineering (ICSE'22). 26% acceptance rate (197/751). 13 pages.
- [C-34] Q. Xu, **J.C. Davis**, Y.C. Hu, and A. Jindal. *An Empirical Study on the Impact of Parameters on Mobile App Energy Usage*. Proceedings of the 29th IEEE International Conference on Software Analysis, Evolution and Reengineering (SANER'22). 36% acceptance rate (72/199). 12 pages.
- [C-35] A. Goel, C. Tung, X. Hu, H. Wang, **J.C. Davis**, Thiruvathukal, and Lu. *Low-Power Multi-Camera Object Re-Identification using Hierarchical Neural Networks*. Proceedings of the ACM/IEEE International Symposium on Low Power Electronics and Design (ISLPED'21). 26.1% acceptance rate (23/88). 6 pages.
- [C-36] **J.C. Davis**, F. Servant, and D. Lee. *Using Selective Memoization to Defeat Regular Expression Denial of Service (ReDoS)*. Proceedings of the 42nd IEEE Symposium on Security and Privacy (IEEE S&P'21). 12% acceptance rate (115/952). 17 pages.
- [C-37] A. Cha, E. Wittern, G. Baudart, **J.C. Davis**, L. Mandel, and J. Laredo. *A Principled Approach to GraphQL Query Cost Analysis*. Proceedings of the 28th ACM Joint European Software Engineering Conference and Symposium on the Foundations of Software Engineering (ESEC/FSE'20). 28% acceptance rate (101/360). 12 pages. *ACM Distinguished Paper Award*.
- [C-38] L. Rupprecht, **J.C. Davis**, C. Arnold, Y. Gur, and D. Bhagwat. *Improving Reproducibility of Data Science Pipelines through Transparent Provenance Capture*. Proceedings of the 46th International Conference on Very Large Data Bases (VLDB'20–Industry track). 20.6% acceptance rate (241/1,172). 15 pages.
- [C-39] **J.C. Davis**, D. Moyer, A. Kazerouni, and D. Lee. *Testing Regex Generalizability And Its Implications: A Large-Scale Many-Language Measurement Study*. Proceedings of the 34th IEEE/ACM International Conference on Automated Software Engineering (ASE'19). 21% acceptance rate (91/435). 13 pages.
- [C-40] L. Michael, J. Donohue, **J.C. Davis**, D. Lee, and F. Servant. *Regexes are Hard: Decision-making, Difficulties, and Risks in Programming Regular Expressions*. Proceedings of the 34th IEEE/ACM International Conference on Automated Software Engineering (ASE'19). 21% acceptance rate (91/435). 12 pages. *ACM Distinguished Paper Award*.
- [C-41] **J.C. Davis**, L. Michael, C. Coghlan, F. Servant, and D. Lee. *Are Regular Expressions a Lingua Franca? An Empirical Study on the Re-use and Portability of Regular Expressions*. Proceedings of the 27th ACM Joint European Software Engineering Conference and Symposium on the Foundations of Software Engineering (ESEC/FSE'19). 24% acceptance rate (97/371). 12 pages.
- [C-42] E. Wittern, A. Cha, **J.C. Davis**, G. Baudart, L. Mandel. *An Empirical Study of GraphQL Schemas*. Proceedings of the 17th International Conference on Service-Oriented Computing (ICSOC'19). 15% acceptance rate (28/183). 16 pages.
- [C-43] X. Fu, T. Ghaffar, **J.C. Davis**, and D. Lee. *EdgeWise: A Better Stream Processing Engine for the Edge*. 2019 USENIX Annual Technical Conference (USENIX ATC'19). 20% acceptance rate (71/356). 17 pages.
- [C-44] **J.C. Davis**, C. Coghlan, F. Servant, and D. Lee. *The Impact of Regular Expression Denial of Service (REDOS) in Practice: an Empirical Study at the Ecosystem Scale*. Proceedings of the 26th ACM Joint European Software Engineering Conference and Symposium on the Foundations of Software Engineering (ESEC/FSE'18). 21% acceptance rate (61/289). 11 pages. *ACM Distinguished Paper Award*.

- [C-45] **J.C. Davis**, E.R. Williamson, and D. Lee. *A Sense of Time for JavaScript and Node.js: First-Class Timeouts as a Cure for Event Handler Poisoning*. Proceedings of the 27th USENIX Security Symposium (USENIX Security'18). 19% acceptance rate (100/520). 17 pages.
- [C-46] **J.C. Davis**, A. Thekumparampil, and D. Lee. *Node.fz: Fuzzing the Server-Side Event-Driven Architecture*. Proceedings of the Twelfth European Conference on Computer Systems (EuroSys'17). 21% acceptance rate (41/200). 16 pages.

## REFEREED JOURNAL ARTICLES<sup>4</sup>

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- [J-1] H. Gao, M. Zahedi, W. Jiang, H.Y. Lin, **J.C. Davis**, and C. Treude. *AI Safety in the Eyes of the Downstream Developer: A First Look at Concerns, Practices, and Challenges*. Empirical Software Engineering, 2026 (EMSE'26). 44 pages.
- [J-2] J. Yasmin, W. Jiang, **J.C. Davis**, and Y. Tian. *Software Dependencies 2.0: An Empirical Study of Reuse and Integration of Pre-Trained Models in Open-Source Projects*. Empirical Software Engineering, 2026 (EMSE'26). 61 pages.
- [J-3] H. Peng, A. Gupte, R. Hasler, N. Eliopoulos, C. Ho, R. Mantri, L. Deng, K. Läuffer, G.K. Thiruvathukal, and **J.C. Davis**. *SysLLMatic: Large Language Models are Software System Optimizers*. Journal of Systems and Software, 2026 (JSS'26). 30 pages.
- [J-4] W. Jiang, M. Kim, C. Cheung, H. Kim, G.K. Thiruvathukal, and **J.C. Davis**. *"I see models being a whole other thing": An Empirical Study of Pre-Trained Model Naming Conventions and A Tool for Enhancing Naming Consistency*. Empirical Software Engineering, 2025 (EMSE'25). 67 pages.
- [J-5] G. Cramer, W. Maxam, and **J.C. Davis**. *Engineering Patterns for Trust and Safety on Social Media Platforms: A Case Study of Mastodon and Diaspora*. Journal of Systems and Software, 2025 (JSS'25). 22 pages.
- [J-6] D. Özkan, K. Davis, **J.C. Davis**, J. Deters, and H. Murzi. *Fostering Systems Thinking through Engineering Study Abroad Programs*. European Journal of Engineering Education, 2024 (EJEE'24). 26 pages.
- [J-7] W. Jiang, V. Banna, N. Vivek, A. Goel, N. Synovic, G.K. Thiruvathukal, and **J.C. Davis**. *Challenges and Practices of Deep Learning Model Reengineering: A Case Study on Computer Vision*. Empirical Software Engineering, 2024 (EMSE'24). 63 pages.
- [J-8] K. Davis, J. Deters, D. Özkan, **J.C. Davis**, and H. Murzi. *Applying Experiential Learning Theory to Understand Study Abroad Leaders' Experiences Using Real-Time Perspectives*. Frontiers: The Interdisciplinary Journal of Study Abroad, Vol. 34, No. 2, 2022 (Frontiers'22). 31 pages.
- [J-9] S. Herbold, A. Trautsch, B. Ledel, A. Aghamohammadi, T.A. Ghaleb, K.K. Chahal, T. Bossenmaier, B. Nagaria, P. Makedonski, M.N. Ahmadabadi, K. Szabados, H. Spieker, M. Madeja, N. Hoy, V. Lenarduzzi, S. Wang, G. Rodriguez-Perez, R. Colomo-Palacios, R. Verdecchia, P. Singh, Y. Qin, D. Chakroborti, W. Davis, V. Walunj, H. Wu, D. Marcilio, O. Alam, A. Aldaej, I. Amit, B. Turhan, S. Eismann, A.K. Wickert, I. Malavolta, M. Sulir, F. Fard, A.Z. Henley, S. Kourtzanidis, E. Tüzün, C. Treude, S.M. Shamasbi, I. Pashchenko, M. Wyrich, **J.C. Davis**, A. Serebrenik, E. Albrecht, E.U. Aktas, D. Strüber, and J. Erbel. *A Fine-grained Data Set and Analysis of Tangling in Bug Fixing Commits*. Empirical Software Engineering, 2022 (EMSE'22). 55 pages.

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<sup>4</sup>In the field of software engineering research, four journals are considered peers of the top conferences, as evidenced by their inclusion in "journal first" tracks at those conferences. those journals are: ACM TOSEM, IEEE TSE, Springer EMSE, and Elsevier JSS.

- [J-10] A. Kazerouni, **J.C. Davis**, A. Basak, C. Shaffer, F. Servant, and S. Edwards. *Fast and Accurate Incremental Feedback for Students' Software Tests Using Selective Mutation Analysis*. Journal of Systems and Software, 2021 (JSS'21). 22 pages.
- [J-11] D. Özkan, K. Davis, **J.C. Davis**, M. James, H. Murzi, and D. Knight. *Expectations and Experiences of Short-Term Study Abroad Leadership Teams*. Journal of International Engineering Education, 2020 (JIEE'20). 34 pages.

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## REFEREED MAGAZINE ARTICLES

- [M-1] T.R. Schorlemmer, E. Burmane, K. Kalu, S. Torres-Arias, and **J.C. Davis**. *Establishing Provenance Before Coding: Traditional and Next-Gen Software Signing*. IEEE Security & Privacy Magazine, special issue "Secure Software Before Codeing", 2025 (IEEE S&P Magazine'25). 8 pages.
- [M-2] X. Hu, Z. Jiao, A. Kocher, Z. Wu, J. Liu, **J.C. Davis**, G.K. Thiruvathukal, and Y.H. Lu. *Evolution of Winning Solutions in the 2021 Low-Power Computer Vision Challenge*. IEEE Computer, 2023 (Computer'23). 6 pages.
- [M-3] A. Goel, C. Tung, N. Eliopoulos, A. Wang, **J.C. Davis**, G.K. Thiruvathukal, and Y.H. Lu. *Tree-based Unidirectional Neural Networks for Low-Power Computer Vision*. IEEE Design & Test, 2022 (IEEE D&T'22). 6 pages.

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## OTHER REFEREED WORKS: VISIONS, TOOLS, PRELIMINARY WORKS, COMPETITIONS

- [W-1] N. Synovic, K. Ryzka, A.V.V. Solari, K. Lyons, **J.C. Davis**, and G.K. Thiruvathukal. *An Empirical Investigation of Pre-Trained Deep Learning Model Reuse in the Scientific Process*. Proceedings of the 22nd IEEE International eScience Conference (short paper) (eScience'26). 6 pages.
- [W-2] T. Singla and **J.C. Davis**. *Software Supply Chains are Dead: Use-Case-Oriented Regeneration*. Proceedings of the 20th International Symposium on Empirical Software Engineering and Measurement – Emerging Results, Vision, and Reflection Track) (ESEM-ERVR'26). 12 pages.
- [W-3] P. Amusuo, D. Liu, R.A.C. Méndez, J. Metzman, O. Chang, and **J.C. Davis**. *Lessons from Mitigating False Positives in Google's OSS-Fuzz-Gen*. Proceedings of the ACM International Conference on the Foundations of Software Engineering (FSE) (FSE-Industry'26). 39% acceptance rate (66/171). 5 pages.
- [W-4] B. Çakar, D. Lee, and **J.C. Davis**. *Towards the Systematic Testing of Regular Expression Engines*. The First Journal Ahead Workshop (co-located with the 2026 International Conference on Software Engineering). Recommended for journal extension; 30/79 papers. (ICSE-JAWs'26). 9 pages.
- [W-5] E. Pautsch\*, T. Singla\*, P. Kumar, W. Jiang, H. Peng, B. Hassanshahi, K. Läufer, G.K. Thiruvathukal, and **J.C. Davis**. *AgentHub: A Registry for Discoverable, Verifiable, and Reproducible AI Agents*. The First Journal Ahead Workshop (co-located with the 2026 International Conference on Software Engineering). Recommended for journal extension; 30/79 papers. (ICSE-JAWs'26). 9 pages.
- [W-6] K.G. Kalu, S. Okorafor, B. Durak, K. Laine, R.C. Moreno, S. Torres-Arias, and **J.C. Davis**. *ARMS: A Vision for Actor Reputation Metric Systems in the Open-Source Software Supply Chain*. The First Journal Ahead Workshop (co-located with the 2026 International Conference on Software Engineering). Recommended for journal extension; 30/79 papers. (ICSE-JAWs'26). 9 pages.

- [W-7] T. Singla, B. Çakar, P.C. Amusuo, and **J.C. Davis**. *Towards a Benchmark for Dependency Decision-Making*. The First Journal Ahead Workshop (co-located with the 2026 International Conference on Software Engineering) (ICSE-JAWs'26). 5 pages.
- [W-8] K. Kalu, S. Rattan, T.R. Schorlemmer, G.K. Thiruvathukal, J.C. Carver, and **J.C. Davis**. *Operationalizing Research Software for Supply Chain Security*. The First Journal Ahead Workshop (co-located with the 2026 International Conference on Software Engineering) (ICSE-JAWs'26). 5 pages.
- [W-9] H. Peng, P.V. Patil, A.Z. Qiu, G.K. Thiruvathukal, and **J.C. Davis**. *Beyond Local Code Optimization: Multi-Agent Reasoning for Software System Optimization*. The First Journal Ahead Workshop (co-located with the 2026 International Conference on Software Engineering) (ICSE-JAWs'26). 5 pages.
- [W-10] H. Peng, A. Zhong, R.A.C. Méndez, K.G. Kalu, and **J.C. Davis**. *How Do Agents Perform Code Optimization? An Empirical Study*. Proceedings of the 23rd International Mining Software Repositories Conference — Mining Challenge track (MSR-Mining'26). 53% acceptance rate (62/116). 5 pages.
- [W-11] P.C. Amusuo, P.V. Patil, O. Cochell, T. Le Lievre, and **J.C. Davis**. *A Unit Proofing Framework for Code-level Verification: A Research Agenda*. Proceedings of the ACM/IEEE 47th International Conference on Software Engineering — New Ideas and Emerging Results track (ICSE-NIER'25). 26% acceptance rate (25/97). 5 pages.
- [W-12] P.V. Patil, W. Jiang, H. Peng, D. Lugo, K.G. Kalu, J. LeBlanc, L. Smith, H. Heo, N. Aou, and **J.C. Davis**. *Recommending Pre-Trained Models for IoT Devices*. Proceedings of the 7th International Workshop on Software Engineering Research & Practices for the Internet of Things (SERP4IoT'25). 5 pages.
- [W-13] S. Joshi, P. Mukherjee, K.A. Davis, and **J.C. Davis**. *Introducing Systems Thinking as a Framework for Teaching and Assessing Threat Modeling Competency*. Proceedings of the 2024 Annual Conference and Exposition of the American Society for Engineering Education (ASEE'24). ~50% acceptance rate (12 papers presented). 31 pages. *Best Paper Award, Software Engineering Division (given to 1 out of 12 papers)*.
- [W-14] B.A. Tanay, L. Arinze, S. Joshi, K.A. Davis, and **J.C. Davis**. *An Exploratory Study on Upper-Level Computing Students' Use of Large Language Models as Tools in a Semester-Long Project*. Proceedings of the 2024 Annual Conference and Exposition of the American Society for Engineering Education (ASEE'24). ~50% acceptance rate (12 papers presented). 27 pages.
- [W-15] **J.C. Davis**, P. Jajal, W. Jiang, T.R. Schorlemmer, N. Synovic, and G.K. Thiruvathukal. *Reusing Deep Learning Models: Challenges and Directions in Software Engineering*. Proceedings of the IEEE John Vincent Atanasoff Symposium on Modern Computing (JVA'23). 12 pages.
- [W-16] J. Srinivasan, S.R. Tanksalkar, P. Amusuo, **J.C. Davis**, and A. Machiry. *Towards Rehosting Embedded Applications as Linux Applications*. Proceedings of the 53rd Annual IEEE/IFIP International Conference on Dependable Systems and Networks — Disrupt track (DSN-Disrupt'23). 47% acceptance rate (17/36). 5 pages.
- [W-17] M. Shen, **J.C. Davis**, and A. Machiry. *Towards Automated Identification of Layering Violations in Embedded Applications (WIP)*. Proceedings of the 24th ACM SIGPLAN/SIGBED International Conference on Languages, Compilers, and Tools for Embedded Systems — Work-In-Progress Track (LCTES-WIP'23). 40% acceptance rate (14/35). 5 pages.
- [W-18] K.G. Kalu, T.R. Schorlemmer, S. Chen, K.A. Robinson, and **J.C. Davis**. *Reflecting on the use of the Policy-Process-Product Theory in Empirical Software Engineering*. Proceedings of the 31st ACM Joint European Software Engineering Conference and Symposium on the Foundations of Software Engineering — Ideas, Visions, and Reflections track (ESEC/FSE-IVR'23). 48% acceptance rate (13/27). 5 pages.

- [W-19] W. Jiang\*, N. Synovic\*, P. Jajal, T.R. Schorlemmer, A. Tewari, B. Pareek, G.K. Thiruvathukal, and **J.C. Davis**. *PTMTorrent: A Dataset for Mining Open-source Pre-trained Model Packages*. Proceedings of the 20th Annual Conference on Mining Software Repositories — Data and Tool Showcase Track (MSR-Data’23). 54% acceptance rate (23/42). 5 pages.
- [W-20] T. Singla, D. Anandayuvraj, K.G. Kalu, T.R. Schorlemmer, and **J.C. Davis**. *An Empirical Study on Using Large Language Models to Analyze Software Supply Chain Security Failures*. Proceedings of the 2nd ACM Workshop on Software Supply Chain Offensive Research and Ecosystem Defenses (SCORED’23). 67% acceptance rate (14/21). 11 pages.
- [W-21] D. Anandayuvraj, P. Thulluri, J. Figueroa, H. Shandilya, and **J.C. Davis**. *Incorporating Failure Knowledge into Design Decisions for IoT Systems: A Controlled Experiment on Novices*. Proceedings of the 5th International Workshop on Software Engineering Research & Practices for the Internet of Things (SERP4IoT’23). 5 pages.
- [W-22] W. Jiang, N. Synovic, R. Sethi, A. Indarapu, M. Hyatt, T.R. Schorlemmer, G.K. Thiruvathukal, and **J.C. Davis**. *An Empirical Study of Artifacts and Security Practices in the Pre-trained Model Supply Chain*. Proceedings of the 1st ACM Workshop on Software Supply Chain Offensive Research and Ecosystem Defenses (SCORED’22). 57% acceptance rate (12/21). 10 pages.
- [W-23] A. Goel, C. Tung, X. Hu, G.K. Thiruvathukal, **J.C. Davis**, and Y.H. Lu. *Efficient Computer Vision on Edge Devices with Pipeline-Parallel Hierarchical Neural Networks*. Proceedings of the 27th Asia and South Pacific Design Automation Conference (ASP-DAC’22). 37% acceptance rate (95/260). 6 pages.
- [W-24] D. Anandayuvraj and **J.C. Davis**. *Reflecting on Recurring Failures in IoT Development*. Proceedings of the 37th IEEE/ACM International Conference on Automated Software Engineering — New Ideas and Emerging Results track (ASE-NIER’22). 36% acceptance rate (18/50). 5 pages.
- [W-25] D. Montes, P. Peerapatanapokin, J. Schultz, C. Guo, W. Jiang, and **J.C. Davis**. *Discrepancies among Pre-trained Deep Neural Networks: A New Threat to Model Zoo Reliability*. Proceedings of the 30th ACM Joint European Software Engineering Conference and Symposium on the Foundations of Software Engineering — Ideas, Visions, and Reflections track (ESEC/FSE-IVR’22). 25% acceptance rate (7/28). 5 pages.
- [W-26] P. Amusuo, A. Sharma, S.R. Rao, A. Vincent, and **J.C. Davis**. *Reflections on Software Failure Analysis*. Proceedings of the 30th ACM Joint European Software Engineering Conference and Symposium on the Foundations of Software Engineering — Ideas, Visions, and Reflections track (ESEC/FSE-IVR’22). 25% acceptance rate (7/28). 6 pages.
- [W-27] C. Okafor\*, T.R. Schorlemmer\*, S. Torres-Arias, and **J.C. Davis**. *SoK: Analysis of Software Supply Chain Security by Establishing Secure Design Properties*. Proceedings of the 1st ACM Workshop on Software Supply Chain Offensive Research and Ecosystem Defenses (SCORED’22). 57% acceptance rate (12/21). 10 pages.
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- [W-29] N. Gopalakrishna, D. Anandayuvraj, A. Detti, F. Bland, S. Rahaman, and **J.C. Davis**. *“If security is required”: Engineering and Security Practices for Machine Learning-based IoT Devices*. Proceedings of the 4th International Workshop on Software Engineering Research & Practices for the Internet of Things (SERP4IoT’22). 8 pages.

- [W-30] **J.C. Davis**, P. Amusuo, and J.R. Bushagour. *Experience Paper: A First Offering of Software Engineering*. Proceedings of the 1st International Workshop on Designing and Running Project-Based Courses in Software Engineering Education (ICSE-DREE'22). 5 pages.
- [W-31] N. Veselsky, J. West, I. Ahlgren, A. Goel, W. Jiang, K. Lee, Y. Kim, **J.C. Davis**, G.K. Thiruvathukal, and N. Klingensmith. *Establishing Trust in Vehicle-to-Vehicle Coordination: A Sensor Fusion Approach*. Proceedings of the 2nd Workshop on Data-Driven and Intelligent Cyber-Physical Systems for Smart Cities (DI-CPS) (DI-CPS'22). 6 pages.
- [W-32] J.M. Winkler, A. Agarwal, C. Tung, D.R. Ugalde, Y.J. Jung, and **J.C. Davis**. *A Replication of "Deep-Bugs: A Learning Approach to Name-based Bug Detection"*. Proceedings of the 29th ACM Joint European Software Engineering Conference and Symposium on the Foundations of Software Engineering (ESEC/FSE'21 Artifact). 1 pages.
- [W-33] **J.C. Davis**. *On the Impact and Defeat of Regex DoS*. ACM Student Research Competition, 2019–2020 Grand Finals. **Second place, graduate student division**.
- [W-34] **J.C. Davis**. *Rethinking Regex Engines to Address ReDoS*. ACM Student Research Competition, 2019–2020 at ESEC/FSE'19. **First place, graduate student division**.
- [W-35] L. Rupprecht, **J.C. Davis**, C. Arnold, A. Lubbock, D. Tyson, and D. Bhagwat. *Ursprung: Provenance for Large-Scale Analytics Environments*. Proceedings of the 2019 International Conference on Management of Data (SIGMOD'19 Demo). 4 pages.
- [W-36] **J.C. Davis**, G. Kildow, and D. Lee. *The Case of the Poisoned Event Handler: Weaknesses in the Node.js Event-Driven Architecture*. Proceedings of the 10th European Workshop on Systems Security (EuroSec'17). 38% acceptance rate (9/24). 6 pages.

## PATENTS

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- [Pa-1] W. Jiang, C. Cheung, H. Kim, M. Kim, and **J.C. Davis**. *A method for identifying naming mismatches in neural networks based on their architectural properties*. Purdue University, U.S. provisional patent application 63/813,549. Filed May 28, 2025.
- [Pa-2] **J.C. Davis** and W. Davis. *Determining a validity of an event emitter based on a rule*. IBM, US Patent 11,875,185 B2. Granted Jan. 16, 2024.
- [Pa-3] W. Davis and **J.C. Davis**. *Verification of the Integrity of Data Files Stored in Copy-on-Write (CoW) Based File System Snapshots*. IBM, U.S. patent 11,176,090 B2. Granted Nov. 16, 2021.
- [Pa-4] **J.C. Davis** and W. Davis. *Injection of Simulated Hardware Failure(s) in a File System for Establishing File System Tolerance-to-Storage-Failure(s)*. IBM, U.S. patent 11,023,341 B2. Granted Jun. 1, 2021.
- [Pa-5] **J.C. Davis**, L. Rupprecht, D. Bhagwat, C. Arnold, and W. Sawdon. *Performing Hierarchical Provenance Collection*. IBM, U.S. patent 10,891,174 B1. Granted Jan. 12, 2021.
- [Pa-6] **J.C. Davis**, W. Davis. *File Metadata Verification in a Distributed File System*. IBM, U.S. patent 10,678,755 B2. Granted Jun. 9, 2020.
- [Pa-7] W. Davis and **J.C. Davis**. *Testing of Lock Managers in Computing Environments*. IBM, U.S. patent 10,061,777 B1. Granted Aug. 28, 2018.
- [Pa-8] **J.C. Davis**, W. Davis, and F. Knop. *Detection of File Corruption in a Distributed File System*. IBM, U.S. patent 10,025,788. Granted Jul. 17, 2018.

## BOOKS AND BOOK CHAPTERS

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- [B-1] **J.C. Davis.** (2026). *Model-Based Agentic Software Engineering*. Book manuscript. <https://davisjam.github.io/model-based-agentic-software-engineering/book/index.html>
- [B-2] **J.C. Davis.** (2023). *Epilogue: The Computer Engineer as Tool-User*. In Y.H. Lu & G.K. Thiruvathukal, *Intermediate C Programming* (2nd edition, pp. 439–442). CRC Press.

## TECHNICAL REPORTS

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- [R-1] K. Kalu, H. Tran, S. Torres-Arias, S. Jeong, and **J.C. Davis.** *A Longitudinal Study of Usability in Identity-Based Software Signing*. <https://arxiv.org/pdf/2603.17133>. 2026.
- [R-2] S. Muralee, S. Cherupattamoolayil, **J.C. Davis**, A. Bianchi, and A. Machiry. *Reactive Bottom-Up Testing*. <https://arxiv.org/pdf/2509.03711>. 2025.
- [R-3] **J.C. Davis**, S. Chen, H. Peng, P. Amusuo, and K. Kalu. *A Guide to Stakeholder Analysis for Cybersecurity Researchers*. <https://arxiv.org/pdf/2508.14796>. 2025.
- [R-4] B. Çakar, C. Sale, S. Chen, E. Burmane, D. Lee, and **J.C. Davis.** *Is Reuse All You Need? A Systematic Comparison of Regular Expression Composition Strategies*. <https://arxiv.org/pdf/2503.20579>. 2025.
- [R-5] N. Ravi, A. Goel, **J.C. Davis**, and G.K. Thiruvathukal. *Improving the Reproducibility of Deep Learning Software: An Initial Investigation through a Case Study Analysis*. <https://arxiv.org/pdf/2505.03165>. 2025.
- [R-6] H. Peng, A. Gupte, N. Eliopoulos, C. Ho, R. Mantri, L. Deng, W. Jiang, Y.H. Lu, K. Läufer, G.K. Thiruvathukal, and **J.C. Davis.** *Large Language Models for Energy-Efficient Code: Emerging Results and Future Directions*. <https://arxiv.org/pdf/2410.09241>. 2024.
- [R-7] V. Purohit, W. Jiang, A.R. Ravikiran, and **J.C. Davis.** *A Partial Replication of MaskFormer in TensorFlow on TPUs for the TensorFlow Model Garden*. <https://arxiv.org/pdf/2404.18801>. 2024.
- [R-8] V. Banna, A. Chinnakotla, Z. Yan, A. Vegesana, N. Vivek, K. Krishnappa, W. Jiang, Y.H. Lu, G.K. Thiruvathukal, and **J.C. Davis.** *An Experience Report on Machine Learning Reproducibility: Guidance for Practitioners and TensorFlow Model Garden Contributors*. <https://arxiv.org/abs/2107.00821>. 2021.

## POSTERS

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- [Ps-1] M. Ahmed, M. Abdelmouty, M. Kim, G. Kandula, A. Park, and **J.C. Davis.** *Advancing Jailbreak Strategies: A Hybrid Approach to Exploiting LLM Vulnerabilities*. Poster Session of the IEEE Secure Development Conference (IEEE SecDev–Poster’25).
- [Ps-2] K.G. Kalu and **J.C. Davis.** *Investigating Software Provenance Consistency in the Open Source Publishing Pipeline*. Poster Session of the 34th USENIX Security Symposium (USENIX Security–Poster’25).
- [Ps-3] B. Çakar, C. Sale, S. Chen, E. Burmane, D. Lee, and **J.C. Davis.** *Is Reuse All You Need? A Systematic Comparison of Regular Expression Composition Strategies*. The 12th Greater Chicago Area Systems Research Workshop (GCASR’25).

- [Ps-4] H. Peng, A. Gupte, R. Hasler, K. L  ufer, G.K. Thiruvathukal, and **J.C. Davis**. *Towards Scalable and Performance-Aware Code Optimization with LLMs*. The 12th Greater Chicago Area Systems Research Workshop (GCASR'25).
- [Ps-5] L. Yadava, P. Kumar, Z. Homrich, R. Potta, P. Kapila, and **J.C. Davis**. *Optimizing Multi-Agent Collaboration in Software Engineering*. The 12th Greater Chicago Area Systems Research Workshop (GCASR'25).
- [Ps-6] K.G. Kalu, S. Torres-Arias, and **J.C. Davis**. *Software Signing: Practical Adoption, Challenges, and Tooling Usability*. 2025 Purdue CERIAS Symposium (CERIAS'25). **Award: Best Poster — 1st-place.**
- [Ps-7] N. Synovic, K. Ryzka, A.V. Solari, **J.C. Davis**, and G.K. Thiruvathukal. *Exploring Deep Neural Network Reuse in Computational Natural Science*. The Loyola University Chicago Undergraduate Research and Engagement Symposium (2025).
- [Ps-8] L. Franke, H. Liang, A. Brantly, **J.C. Davis**, and C. Brown. *A First Look at the General Data Protection Regulation (GDPR) in Open-Source Software*. Proceedings of the ACM/IEEE 46th International Conference on Software Engineering — Poster Track (ICSE-Poster'24).
- [Ps-9] S.R. Tanksalkar, J. Srinivasan, S. Danduri, P. Amusuo, **J.C. Davis**, and A. Machiry. *LeMix: Rehosting Embedded Systems as Linux Applications for Effective Vulnerability Detection*. 2024 Purdue CERIAS Symposium (CERIAS'24). **Award: Best Poster — 1st-place.**
- [Ps-10] K.G. Kalu, S. Torres-Arias, and **J.C. Davis**. *Navigating Software Supply Chain Risks: Practitioner Perspectives on Software Signing*. 2024 Purdue CERIAS Symposium (CERIAS'24).
- [Ps-11] T.R. Schorlemmer, W. Jiang, and **J.C. Davis**. *Machine Learning Supply Chain Security*. 2023 Purdue CERIAS Symposium (CERIAS'23). **Award: Best Poster — 2nd-place.**
- [Ps-12] W. Jiang, T.R. Schorlemmer, and **J.C. Davis**. *Trustworthy Re-use of Pre-trained Neural Networks*. 2023 Purdue CERIAS Symposium (CERIAS'23).
- [Ps-13] W. Maxam and **J.C. Davis**. *Plan for an evaluation of government cyber threat hunting processes*. 2022 Purdue CERIAS Symposium (CERIAS'22).
- [Ps-14] N. Hornbrook and **J.C. Davis**. *An Intercultural Engineering Module for Software Engineers*. 2021 Annual Colloquium for International Engineering Education (ACIEE'21).
- [Ps-15] N. Vivek, A. Chinnakotla, V. Banna, A. Vegesana, Z. Yan, **J.C. Davis**, Y.H. Lu, and G.K. Thiruvathukal. *Exemplars for Machine Learning: Towards Software Engineering & Reproducibility*. SIAM Conference on Computational Science and Engineering (CSE'21).

## COURSES DESIGNED\* OR RE-DESIGNED†

|                                                               |                      |
|---------------------------------------------------------------|----------------------|
| †ECE 461 — Software Engineering<br>Purdue University          | Launched Fall 2021   |
| *ECE 595 — Advanced Software Engineering<br>Purdue University | Launched Spring 2021 |
| †ECE 30862 — Software Engineering Tools<br>Purdue University  | Revamped Fall 2021   |

## COURSES TAUGHT

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|----------------------------------------------------------------------------------------------------------|-----------------------------------|
| <b>ECE 461 – Software Engineering</b><br><i>Purdue University</i>                                        | F21, S23, F23, F24, F25           |
| <b>ECE 595 – Advanced Software Engineering</b><br><i>Purdue University</i>                               | S21, S22, S24, S26                |
| <b>ECE 368 – Data Structures</b><br><i>Purdue University</i>                                             | Fall 2020                         |
| <b>Vertically Integr. Project: Software Engineering w/Pre-Trained Models</b><br><i>Purdue University</i> | S24, F24, S25, F25, S26           |
| <b>Vertically Integr. Project: Open-Source TensorFlow Software</b><br><i>Purdue University</i>           | F20, S21, F21, S22, F22, S23, F23 |
| <b>Vertically Integr. Project: SafeRegex</b><br><i>Purdue University</i>                                 | Fall 2020, Spring 2021            |
| <b>CS 3114 – Data Structures and Algorithms</b><br><i>Virginia Tech</i>                                  | Fall 2019                         |
| <b>CS 1064 – Introduction to Programming in Python</b><br><i>Virginia Tech</i>                           | Spring 2019                       |
| <b>Rising Sophomore Abroad Program (Track Leader)</b><br><i>Virginia Tech</i>                            | Spring 2018, Spring 2019          |

## GRADUATE STUDENT ADVISING

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### DOCTORAL STUDENTS, AS COMMITTEE CHAIR

|                                       |               |                                           |
|---------------------------------------|---------------|-------------------------------------------|
| Dr. Wenxin Jiang                      | PhD           | <i>Graduated 2025</i>                     |
| Dr. Paschal Amusuo                    | PhD           | <i>Graduated 2026</i>                     |
| Dharun Anandayuvraj                   | PhD candidate | Fall 2021–present, expected Fall 2026     |
| Purvish Jajal (with Y.H. Lu)          | PhD candidate | Fall 2022–present, expected Fall 2027     |
| Nicholas J. Eliopoulos (with Y.H. Lu) | PhD candidate | Fall 2024–present, expected Spring 2027   |
| Kelechi Gabriel Kalu                  | PhD candidate | Spring 2023–present, expected Spring 2027 |
| Daniel Lugo, US Space Force           | PhD student   | Fall 2024–present, expected Spring 2027   |
| Andrew Rozema                         | PhD student   | Fall 2024–present, expected Spring 2028   |
| Berk Çakar                            | PhD student   | Fall 2024–present, expected Spring 2029   |
| Huiyun Peng                           | PhD student   | Fall 2024–present, expected Spring 2029   |
| Ricardo Andrés Calvo Méndez           | PhD student   | Fall 2025–present, expected Spring 2030   |

### MSC-THESIS STUDENTS, AS COMMITTEE CHAIR

|                         |     |                       |
|-------------------------|-----|-----------------------|
| Tanmay Singla           | MSc | <i>Graduated 2026</i> |
| Sofia Okorafor, US Navy | MSc | <i>Graduated 2026</i> |
| Parth V. Patil          | MSc | <i>Graduated 2025</i> |
| Jason Jones             | MSc | <i>Graduated 2024</i> |

|                               |     |                       |
|-------------------------------|-----|-----------------------|
| Taylor Schorlemmer, US Army   | MSc | <i>Graduated 2024</i> |
| William Maxam, US Coast Guard | MSc | <i>Graduated 2023</i> |
| Geoffrey Cramer               | MSc | <i>Graduated 2023</i> |

#### DOCTORAL STUDENTS, AS COMMITTEE MEMBER

|                  |     |                       |
|------------------|-----|-----------------------|
| Dr. Qiang Xu     | PhD | <i>Graduated 2024</i> |
| Dr. Abhinav Goel | PhD | <i>Graduated 2022</i> |

|                                      |               |                     |
|--------------------------------------|---------------|---------------------|
| Andreas Kellas (Columbia University) | PhD candidate | Spring 2025–present |
| Benjamin Shiue-Hal Chou              | PhD candidate | Fall 2023–present   |
| Chinenye Okafor                      | PhD candidate | Fall 2022–present   |

|                       |             |                   |
|-----------------------|-------------|-------------------|
| Sai Ritvik Tanksalkar | PhD student | Fall 2022–present |
| Akul Pillai           | PhD student | Fall 2022–present |
| Mingjie Shen          | PhD student | Fall 2021–present |

#### MSC-THESIS STUDENTS, AS COMMITTEE MEMBER

|                      |     |                       |
|----------------------|-----|-----------------------|
| Jayashree Srinivasan | MSc | <i>Graduated 2023</i> |
| Xiao Hu              | MSc | <i>Graduated 2022</i> |

#### STUDENT AWARDS AND FELLOWSHIPS

*These are students I mentored.*

#### NATIONAL AND INTERNATIONAL AWARDS

|                                                                           |      |
|---------------------------------------------------------------------------|------|
| Huiyun Peng, “Noteworthy Reviewer” for USENIX Security 2026—Artifact PC   | 2026 |
| Paschal C. Amusuo & Dharun Anandayuvraj, Qualcomm Innovation Fellows      | 2025 |
| Evan Williams, CSGrad4US Fellow                                           | 2025 |
| Arjun Gupte, Astronaut Scholar                                            | 2025 |
| Dharun Anandayuvraj, Chateaubriand Fellow                                 | 2024 |
| Matthew Hyatt, DOD NDSEG Fellow; NSF Graduate Research Fellow (GRFP)      | 2023 |
| Vishnu Banna, Honorable Mention in CRA Undergraduate Research Competition | 2021 |
| Vishnu Banna, Google Open Source Peer Bonus                               | 2021 |
| Anirudh Vegesana, Google Open Source Peer Bonus                           | 2021 |
| Akhil Chinnakotla, Google Open Source Peer Bonus                          | 2021 |

## INSTITUTIONAL AWARDS

|                                                                                              |      |
|----------------------------------------------------------------------------------------------|------|
| VIP Team: Presentation with Distinction Award, 2026 Spring Undergraduate Research Conference | 2026 |
| Arjun Gupte, VIP Outstanding Leadership Award                                                | 2026 |
| Paschal C. Amusuo, CERIAS Diamond Award                                                      | 2026 |
| Paschal C. Amusuo, ECE Magoon Graduate Student Research Excellence Award                     | 2026 |
| Arjun Gupte, ECE Undergraduate Research Award                                                | 2026 |
| Elijah Jorgensen, Rolls Royce Fellow                                                         | 2026 |
| Wenxin Jiang, ECE Magoon Graduate Student Research Excellence Award                          | 2025 |
| Jason Jones, ECE Magoon Graduate Student Teaching Excellence Award                           | 2024 |
| Paschal C. Amusuo, ECE Magoon Graduate Student Teaching Excellence Award                     | 2022 |
| Kyle Robinson, ECE Undergraduate Research Award                                              | 2024 |
| Efe Barlas, ECE Undergraduate Research Award                                                 | 2022 |
| Vishnu Banna, ECE Undergraduate Research Award                                               | 2021 |
| Joseph Bushagour, ECE Undergraduate Excellence Award for Teaching                            | 2022 |
| Joseph Thomas, ECE Magoon Graduate Student Teaching Excellence Award                         | 2021 |
| Venkat Manam, ECE Magoon Graduate Student Teaching Excellence Award                          | 2021 |

## INVITED TALKS

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### ON MODEL-BASED AGENTIC SOFTWARE ENGINEERING (MAGE)

|                                                                             |      |
|-----------------------------------------------------------------------------|------|
| U.S. Space Force. “Brain Day”, unit Systems Delta (SYD) 85                  | 2026 |
| IBM T.J. Watson Research Center. Yorktown Heights, NY                       | 2026 |
| Argonne National Laboratories. Lemont, IL                                   | 2026 |
| Purdue University, College of Engineering. ThinkAI Summer Workshop Series   | 2026 |
| Purdue University, ECE. Workshop on “ <i>Job Interviews in the AI-Era</i> ” | 2026 |

### ON SOFTWARE ENGINEERING FOR MACHINE LEARNING — REUSING PRE-TRAINED MODELS

|                                                             |      |
|-------------------------------------------------------------|------|
| Oregon State University. Corvallis, OR                      | 2025 |
| McGill University. Montreal, Canada                         | 2025 |
| Concordia University. Montreal, Canada                      | 2025 |
| The University of Arizona. Tucson, AZ                       | 2024 |
| Michigan Technological University. Houghton, MI             | 2024 |
| Carnegie Mellon University. Pittsburgh, PA                  | 2023 |
| The University of Notre Dame. South Bend, IN                | 2023 |
| Argonne National Laboratories. Lemont, IL                   | 2023 |
| ONNX Community Meetup, NVIDIA Headquarters. Santa Clara, CA | 2023 |
| Loyola University Chicago. Chicago, IL                      | 2022 |

### ON REGULAR EXPRESSION DENIAL OF SERVICE (REDOS)

|                                                                  |      |
|------------------------------------------------------------------|------|
| Dagstuhl seminar on “Regular Expressions: Matching and Indexing” | 2024 |
|------------------------------------------------------------------|------|

|                                                  |      |
|--------------------------------------------------|------|
| Clemson University. Clemson, SC                  | 2020 |
| Clarkson University. Potsdam, NY                 | 2020 |
| Colorado School of Mines. Golden, CO             | 2020 |
| Pennsylvania State University. State College, PA | 2020 |
| University of Nebraska. Lincoln, NE              | 2020 |
| York University. Toronto, Canada                 | 2019 |

## ON SOFTWARE ENGINEERING FOR EMBEDDED SYSTEMS

|                                                                                                                                                                                       |      |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------|
| Securing Software Implementations through System Fuzz Testing and Modular Formal Methods<br><i>Rolls Royce Annual Conference, Cyber Technology Research Network. Indianapolis, IN</i> | 2025 |
| Quick and Dirty or Slow and Steady? Two Techniques for Validating Embedded Software<br><i>Rolls Royce Headquarters. Derby, UK</i>                                                     | 2024 |

## OTHER TALKS

|                                                                                                                                                                                                    |                   |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------|
| What Can We Learn from Failure in Software Engineering?<br><i>Engineering Innovations: The Official Podcast of Purdue University's Elmore Family School of Electrical and Computer Engineering</i> | 2025              |
| Failure-Aware SW Development Lifecycles: Opportunities for Intra-/Inter-Org. Learning<br><i>Purdue CERIAS External Advisory Board</i>                                                              | 2024              |
| An Interview Study on 3 <sup>rd</sup> -Party Cyber Threat Hunting Processes in the U.S. Dep't of Homeland Sec.<br><i>Purdue CERIAS Annual Symposium</i>                                            | 2024              |
| Challenges in Global Software Development<br><i>University of Wisconsin-Stout. Menomonie, WI</i>                                                                                                   | 2021              |
| Regexes are Hard: Qualitative and Quantitative Perspectives<br><i>North Carolina State University. Raleigh, NC</i>                                                                                 | 2019              |
| Regexes in the Wild<br><i>Virginia Tech. Blacksburg, VA</i>                                                                                                                                        | 2019              |
| The Dangers of Copy/Pasting Code<br><i>Episode of the Podcast "The Secure Developer"</i>                                                                                                           | 2019              |
| Academic Perspectives on Node.js<br><i>Node.js Collaborator Summit. Vancouver, Canada</i>                                                                                                          | 2018              |
| International Engineering<br><i>Rising Sophomore Abroad Program, Virginia Tech. Blacksburg, VA</i>                                                                                                 | Annual, 2015–2019 |

## ACADEMIC COMMUNITY SERVICE

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### REVIEWER: MAJOR CONFERENCE TECHNICAL TRACKS<sup>5</sup>

|                             |                  |
|-----------------------------|------------------|
| Vice Chair, USENIX Security | 2027             |
| PC Member, ICSE             | 2025, 2026, 2027 |
| PC Member, AAAI             | 2026, 2027       |
| PC Member, USENIX Security  | 2025, 2026       |
| PC Member, ESEC/FSE         | 2023, 2025       |
| PC Member, ASE              | 2021, 2024       |
| PC Member, ISSTA            | 2024, 2025       |

### REVIEWER: JOURNALS

|                                                                            |
|----------------------------------------------------------------------------|
| Reviewer, IEEE Transactions on Software Engineering (TSE)                  |
| Reviewer, ACM Transactions on Software Engineering and Methodology (TOSEM) |
| Reviewer, Springer Empirical Software Engineering (EMSE)                   |
| Reviewer, Journal of Systems and Software (JSS)                            |
| Reviewer, Journal of Online Trust & Safety (JOTS)                          |

### REVIEWER: OTHER SERVICE AS REFEREE (MINOR VENUES OR NON-TECHNICAL TRACKS)

|                                                                                         |                  |
|-----------------------------------------------------------------------------------------|------------------|
| PC Member, MSR Mining Challenge                                                         | 2026             |
| PC Member, European Workshop on Systems Security (EuroSec)                              | 2024, 2025, 2026 |
| PC Member, IEEE SecDev                                                                  | 2024             |
| Reviewer, IEEE-CS SWEBOK Guide V4 (Guide to the Software Engineering Body of Knowledge) | 2024             |
| Reviewer, SANER–Early Research Achievement Track                                        | 2024             |
| PC Member, Twelfth Workshop on Education for High-Performance Computing (EduHPC)        | 2024             |
| PC Member, LCTES                                                                        | 2023             |
| Reviewer, SCAM–Engineering track                                                        | 2023             |
| PC Member, ACM Workshop on Software Supply Chain Offens. Res. and Ecosystem Def.        | 2022, 2023       |
| Reviewer, ASE–Doctoral Symposium Track                                                  | 2022             |
| Reviewer, ICSE–Demonstrations Track                                                     | 2021             |
| Reviewer, ESEC/FSE–Artifact Track                                                       | 2020, 2021       |
| Judge, CSAW’21 Best Paper Competition                                                   | 2021, 2023       |
| Reviewer, CGO–Artifact Track                                                            | 2019             |
| Sub-reviewer: Middleware’17, ASPLOS’18, EuroSys’18, MASCOTS’18, HPCA’19, CGO’19         | 2016–2019        |

### WORKSHOP AND EVENT ORGANIZATION

|                                                                                    |           |
|------------------------------------------------------------------------------------|-----------|
| Co-Chair, Workshop on Research Software Supply Chain Security (RS3), IEEE eScience | 2026      |
| Organizer, Workshop on Using Generative AI as an Engineering Educator, ASEE 2026   | 2026      |
| Co-Chair, ICSE Student Mentoring Workshop (ICSE-SMeW)                              | 2024–2025 |

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<sup>5</sup>In my research areas, service on the technical program committees (“PC member”) is a respected form of peer review service. This involves the review of 5–15 full papers per conference.

|                                                                  |      |
|------------------------------------------------------------------|------|
| Panelist and Mentor, ICSE Student Mentoring Workshop (ICSE-SMeW) | 2024 |
| Mentor, ICSE Student Mentoring Workshop (ICSE-SMeW)              | 2023 |

## NATIONAL SERVICE

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|                                          |      |
|------------------------------------------|------|
| US National Science Foundation, Panelist | 2025 |
| US National Science Foundation, Panelist | 2023 |

## UNIVERSITY SERVICE

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|                                                                                         |              |
|-----------------------------------------------------------------------------------------|--------------|
| ECE Representative on the COE Graduate Education Award Selection Committee              | 2025         |
| Lead for BSc.-Computer Engineering, Purdue ECE ABET Committee 5-year self-evaluation    | 2024–2025    |
| Organizer, Junior Faculty Peer Mentoring Teatime                                        | 2024–2025    |
| Organizer, CAREER writing group                                                         | 2024, 2025   |
| Organizer, Software Systems Reading Group                                               | 2024         |
| Member, Committee to Create MSc-Software Engineering                                    | 2024         |
| Breakout session discussion lead: LLMs in education — ECE ADVANCE                       | 2024         |
| Member, Purdue ECE Faculty Search Committee – Prof. of Practice in Software Engineering | 2023–2024    |
| Member, Purdue ECE ABET Committee                                                       | 2023–present |
| Member, Purdue ECE Ad Hoc Faculty Search Committee                                      | 2023         |
| Member, Purdue ECE Faculty Search Committee — Software Engineering                      | 2022–2023    |
| Host, Computer Engineering Seminar Series — Dr. Joanna C. S. Santos (Notre Dame)        | 2022         |
| Host, Purdue Engineering Distinguished Lecture Series (PEDLS) — Dr. Nancy Leveson (MIT) | 2022         |
| Member, Purdue ECE Undergraduate Curriculum Committee                                   | 2020–2022    |
| Panelist, CS@Virginia Tech Academic Jobs Panel                                          | 2021         |
| President, Virginia Tech CS Graduate Student Council                                    | 2018–2019    |
| Organizer, Virginia Tech Systems Reading Group                                          | 2017–2020    |

## SHORT COURSES AND WORKSHOPS ATTENDED

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|                                                                                                                       |      |
|-----------------------------------------------------------------------------------------------------------------------|------|
| Dagstuhl seminar on “Regular Expressions: Matching and Indexing”                                                      | 2024 |
| Generative AI Assistance in Grant Proposal Writing (Purdue University–Internal)                                       | 2024 |
| Inclusive Research as a Pathway to Broadening Participation and Instit. Excellence (NSF+ODIB)                         | 2024 |
| Leadership Skills for Engineering and Science Faculty (Leiserson and McVinney)                                        | 2024 |
| NSF Grand Challenges in Resilience Workshop, Purdue University                                                        | 2023 |
| Tools to Foster Students’ (Cross-)cultural Sensitivity in Engineering Ethical Decision-Making (ASEE’22, Clancy & Qiu) | 2022 |
| Effective College Teaching (Brent & Felder)                                                                           | 2020 |
| Intercultural Pedagogy Grant Training Program, Purdue CILMAR                                                          | 2020 |

## PROFESSIONAL MEMBERSHIPS

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Senior Member, Institute of Electrical and Electronics Engineers (IEEE)  
Member, Association for Computing Machinery (ACM)  
Member, American Society for Engineering Education (ASEE)

## AWARDS AND RECOGNITION

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### FOR RESEARCH

|                                                                                            |           |
|--------------------------------------------------------------------------------------------|-----------|
| NSF CAREER Award                                                                           | 2026      |
| Paul C. Krause Faculty Research Award ( <i>ECE@Purdue Junior Faculty Research</i> )        | 2025      |
| Best Artifact Award, CCS'25 (PickleBall)                                                   | 2025      |
| Best Paper Award, Software Engineering Division, ASEE 2024 ( <i>Educational research</i> ) | 2024      |
| ACM Distinguished Paper Award, ESEC/FSE 2020                                               | 2020      |
| Second place, Grand Finals of the ACM Graduate Student Research Competition                | 2020      |
| First place, Graduate Student Research Competition, ESEC/FSE 2019                          | 2019      |
| ACM Distinguished Paper Award, ASE 2019                                                    | 2019      |
| Davenport Fellowship, Virginia Tech College of Engineering                                 | 2019      |
| Graduate Fellow, VT Academy for Global Engineering                                         | 2019–2020 |
| Microsoft Security Researcher Acknowledgments ( <i>for Regex DoS discoveries</i> )         | 2018      |
| IBM Significant Contributor Award ( <i>for Node.js</i> )                                   | 2018      |
| ACM Distinguished Paper Award, ESEC/FSE 2018                                               | 2018      |
| Pratt Fellowship, Virginia Tech College of Engineering                                     | 2017–2019 |

### FOR TEACHING

|                                                                                             |                        |
|---------------------------------------------------------------------------------------------|------------------------|
| Purdue University Exceptional Early Career Teaching Award                                   | 2026                   |
| Purdue College of Engineering Faculty Excellence Award in Exceptional Early Career Teaching | 2026                   |
| 2022 Ruth and Joel Spira Outstanding Teacher Award                                          | 2022                   |
| “Outstanding Engineering Teachers” (COE) ( <i>course evaluation scores</i> )                | F'21, S'22, F'22, S'24 |

### FOR SERVICE

|                                                   |      |
|---------------------------------------------------|------|
| ICSE 2026 Distinguished Reviewer Award            | 2026 |
| FSE 2025 Distinguished Reviewer Award             | 2025 |
| USENIX 2025 Distinguished Reviewer Award          | 2025 |
| ASE 2021 Distinguished PC Member Award            | 2021 |
| Outstanding Graduate Student Service Award, CS@VT | 2020 |

### FOR MENTORING

|                                                                                          |      |
|------------------------------------------------------------------------------------------|------|
| VIP Outstanding Team Mentor Award, Purdue TensorFlow Team                                | 2026 |
| Outstanding Faculty Mentor — Purdue's Elmore School of Electrical & Computer Engineering | 2024 |
| VIP Outstanding Team Mentor Award, Purdue TensorFlow Team                                | 2021 |

### OTHER HONORS

|                                                                                             |      |
|---------------------------------------------------------------------------------------------|------|
| Purdue ThinkAI Competition Winner ( <i>COE-wide competition for creative use of GenAI</i> ) | 2026 |
| Frederica Clarkson Award ( <i>university-wide award for top graduating senior</i> )         | 2012 |